

TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

NATIONAL COLLEGE OF ENGINEERING

A

CASE STUDY REPORT

ON

NETWORK TOPOLOGY DESIGN FOR TRIBHUVAN INTERNATIONAL AIRPORT

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Abbreviations

ACL	Access Control List
AP	Access Point
ATC	Air Traffic Control
CAAN	Civil Aviation Authority of Nepal
CNS	Communication, Navigation, and Surveillance
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
GW	Default Gateway
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ICAO	International Civil Aviation Organization
IP	Internet Protocol
ISP	Internet Service Provider
LAN	Local Area Network
MAC	Media Access Control
OSPF	Open Shortest Path First
SSID	Service Set Identifier
TETRA	Terrestrial Trunked Radio
TIA	Tribhuvan International Airport
VLAN	Virtual Local Area Network
VNKT	ICAO code for Tribhuvan International Airport
WAN	Wide Area Network
WPA2	Wi-Fi Protected Access 2

1 Objectives

(i) To design and implement a segmented dual-terminal airport network using VLANs, Router-on-a-Stick, and OSPF in Cisco Packet Tracer.

(ii) To configure dedicated Server-PT DHCP servers with ip helper-address relay, DNS, and HTTP services for both terminals.

(iii) To implement zero-trust guest isolation using Extended Named ACLs while still permitting guest access to a public airport information portal.

2 Theory

2.1 Switch

Switches build the local network inside each terminal and forward traffic at Layer 2. Cisco 2960-24TT switches were used because they support VLANs, access ports, and 802.1Q trunking in Packet Tracer. Each terminal has one core switch and several edge switches, so devices stay organized and easy to troubleshoot. By placing ports into the correct VLAN, the switches give the first level of separation between staff, airline, guest, and operational users.

2.2 Router

Routers connect the VLANs, the inter-terminal backbone, and the ISP links. TIA-R1 is the gateway for the International terminal and TIA-R2 is the gateway for the Domestic terminal. Cisco 2911 routers were selected because they support subinterfaces, OSPF, ACLs, and DHCP relay in Packet Tracer. In this project, the routers provide default gateways, relay DHCP requests, exchange routes, and enforce guest and management security policies.

2.3 VLAN

VLANs divide one physical switch network into separate logical networks. This is important in an airport because guests, airline staff, immigration staff, and operational systems should not share the same broadcast domain. The design uses six VLANs: 10, 20, 30, 40, 50, and 60, each with its own /24 subnet. That separation makes address-

ing, DHCP, routing, and ACL policies easier to control and also reduces unnecessary broadcast traffic.

2.4 Router-on-a-Stick

Router-on-a-Stick is used for inter-VLAN routing. A single router interface connects to the core switch as a trunk, and separate subinterfaces act as the gateways for different VLANs. TIA-R1 uses subinterfaces for VLANs 10, 20, and 30, while TIA-R2 uses subinterfaces for VLANs 40, 50, and 60. This gives the project simple inter-VLAN routing without needing a Layer-3 switch.

2.5 OSPF

OSPF is the routing protocol between TIA-R1 and TIA-R2. The routers form a neighbor relationship over the 10.0.0.0/30 backbone and exchange routes in Area 0. Each router advertises its local VLANs and WAN segment, while its static default route is shared with `default-information originate`. This avoids manual route entry and lets each terminal learn the other side automatically.

2.6 DHCP

DHCP gives clients their IP address, subnet mask, gateway, and DNS values automatically. In this network, DHCP is provided by two Server-PT devices, not by the routers. DHCP-Intl-Server serves VLANs 10, 20, and 30, while DHCP-Dom-Server serves VLANs 40, 50, and 60. The routers use `ip helper-address` on all user-facing subinterfaces so clients can reach the correct DHCP server even when it is in another subnet.

2.7 DNS

DNS lets users reach airport services by name instead of by IP address. Both DNS servers store the same two A records: `tia.caan.gov.np` for the staff portal at 192.168.10.100 and guest.tia.caan.gov.np for the guest portal at 192.168.40.2. Keeping a DNS server on each side gives the Domestic terminal a local resolver while still keeping name resolution consistent across the whole airport.

2.8 HTTP

HTTP is the visible end service in this project. The staff portal runs on CAAN-Server at 192.168.10.100, and the guest portal runs on DNS-Dom-Server at 192.168.40.2. When a page loads, it proves that DHCP, DNS, routing, and ACL policy are all working together. Using different server IPs for staff and guest web access also makes the guest security rule easier to enforce.

2.9 Static Routing

Static routing is used where it is the simplest choice. TIA-R1 points its default route to NT-Router at 203.0.113.1, and TIA-R2 points its default route to WL-Router at 198.51.100.1. The ISP routers do not run OSPF; they only keep static return routes for the six airport VLANs. This keeps the provider side simple and clearly separate from the airport routing domain.

2.10 Extended ACLs

Extended ACLs control guest traffic more precisely than standard ACLs because they can match source, destination, protocol, and port. INTL_GUEST_SECURE is applied on VLAN 30 and DOM_GUEST_SECURE is applied on VLAN 50. These rules allow only approved DNS traffic and web access to the public guest portal, then block access to private airport networks. A separate MGMT_ONLY ACL protects router login access on the VTY lines.

3 Network Topology

The TIA network uses two star-shaped terminal blocks. On the International side, TIA-R1 connects to Sw-Intl-Core, which then connects to edge switches for Immigration, International Airlines, and International Guest Wi-Fi. On the Domestic side, TIA-R2 connects to Sw-Dom-Core, which connects to edge switches for Domestic Airlines, Domestic Guest Wi-Fi, and ATC/TETRA CNS. This keeps each terminal easy to read while still giving every user group its own VLAN and subnet.

The two terminal routers are joined by the 10.0.0.0/30 backbone, where OSPF exchanges internal routes. TIA-R1 connects to NT-Router on 203.0.113.0/30 and TIA-R2 connects to WL-Router on 198.51.100.0/30. The International side hosts the staff portal, primary DNS, and International DHCP server, while the Domestic side keeps its own DNS and DHCP server and hosts the public guest portal. This topology supports routing, DHCP relay, DNS, web access, and guest isolation without adding technologies outside the project scope.

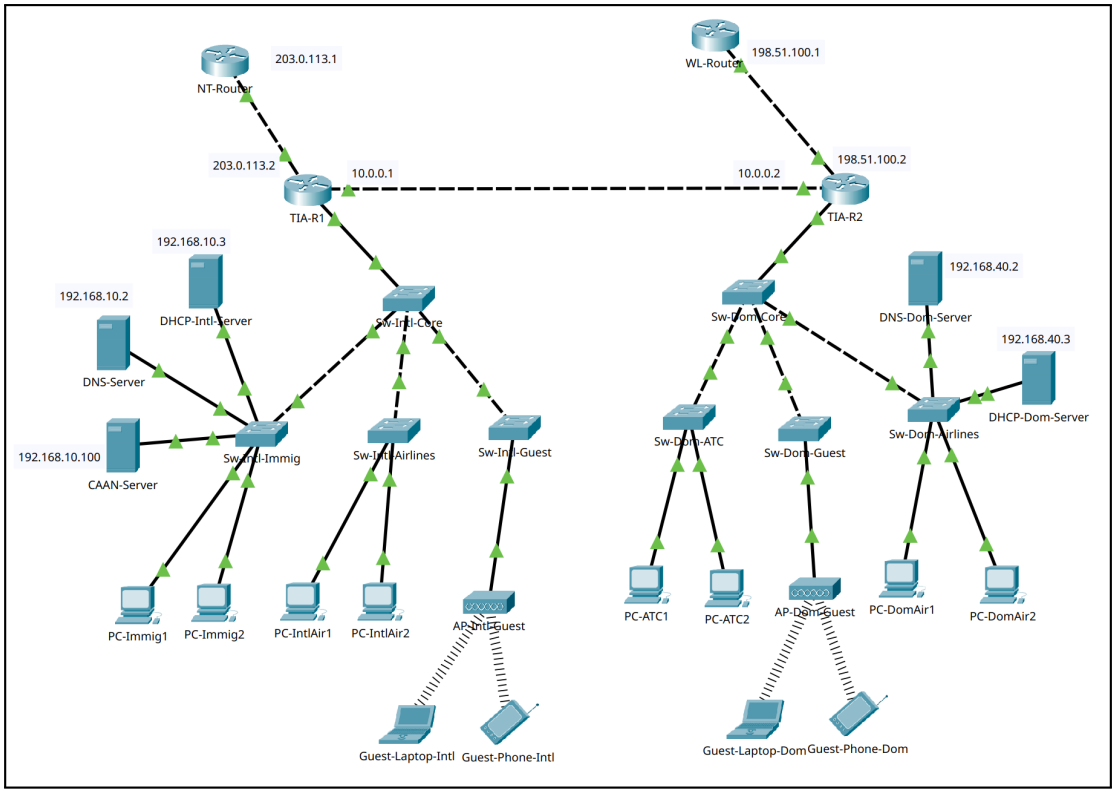


Figure 1: Complete dual-terminal airport topology in Cisco Packet Tracer.

4 Description

4.1 VLAN Design

Table 1: VLAN design used in the TIA network

VLAN	Name	Subnet	Purpose
10	Immigration and Authority	192.168.10.0/24	Staff and administrative systems on the International side.
20	International Airlines	192.168.20.0/24	International airline office devices.
30	International Guest Wi-Fi	192.168.30.0/24	Public wireless users in the International Terminal.
40	Domestic Airlines	192.168.40.0/24	Domestic airline office devices.
50	Domestic Guest Wi-Fi	192.168.50.0/24	Public wireless users in the Domestic Terminal.
60	ATC and TETRA CNS	192.168.60.0/24	Sensitive airport operational systems.

The VLAN plan separates users by role, so public, staff, airline, and operational traffic stay apart from the start. This improves security, reduces broadcasts, and makes policy enforcement easier.

4.2 IP Addressing and DHCP

Table 2: IP address plan

Segment	Subnet	Gateway	Use
VLAN 10	192.168.10.0/24	192.168.10.1	Immigration and Authority
VLAN 20	192.168.20.0/24	192.168.20.1	International Airlines
VLAN 30	192.168.30.0/24	192.168.30.1	International Guest Wi-Fi
VLAN 40	192.168.40.0/24	192.168.40.1	Domestic Airlines
VLAN 50	192.168.50.0/24	192.168.50.1	Domestic Guest Wi-Fi
VLAN 60	192.168.60.0/24	192.168.60.1	ATC and TETRA CNS
R1-R2 Backbone	10.0.0.0/30	N/A	Inter-terminal link
R1-NT WAN	203.0.113.0/30	N/A	Nepal Telecom uplink
R2-WL WAN	198.51.100.0/30	N/A	Worldlink uplink

Table 3: Router interface addresses

Device	Interface	IP address
TIA-R1	GigabitEthernet0/0.10	192.168.10.1/24
TIA-R1	GigabitEthernet0/0.20	192.168.20.1/24
TIA-R1	GigabitEthernet0/0.30	192.168.30.1/24
TIA-R1	GigabitEthernet0/1	10.0.0.1/30
TIA-R1	GigabitEthernet0/2	203.0.113.2/30
TIA-R2	GigabitEthernet0/0.40	192.168.40.1/24
TIA-R2	GigabitEthernet0/0.50	192.168.50.1/24
TIA-R2	GigabitEthernet0/0.60	192.168.60.1/24
TIA-R2	GigabitEthernet0/1	10.0.0.2/30
TIA-R2	GigabitEthernet0/2	198.51.100.2/30

Table 4: Static IP addresses for servers

Server	IP address	Role
CAAN-Server	192.168.10.100	Hosts the internal staff portal
DNS-Server	192.168.10.2	Main DNS server
DNS-Dom-Server	192.168.40.2	Domestic-side DNS server and public guest web host
DHCP-Intl-Server	192.168.10.3	DHCP for VLANs 10, 20, and 30
DHCP-Dom-Server	192.168.40.3	DHCP for VLANs 40, 50, and 60

Table 5: DHCP pool summary configured on the Server-PT devices

Server	Pool	Gateway	DNS	Range
DHCP-Intl-Server	VLAN 10	192.168.10.1	192.168.10.2	192.168.10.11 to 192.168.10.254
DHCP-Intl-Server	VLAN 20	192.168.20.1	192.168.10.2	192.168.20.11 to 192.168.20.254
DHCP-Intl-Server	VLAN 30	192.168.30.1	192.168.10.2	192.168.30.11 to 192.168.30.254
DHCP-Dom-Server	VLAN 40	192.168.40.1	192.168.40.2	192.168.40.11 to 192.168.40.254
DHCP-Dom-Server	VLAN 50	192.168.50.1	192.168.40.2	192.168.50.11 to 192.168.50.254
DHCP-Dom-Server	VLAN 60	192.168.60.1	192.168.40.2	192.168.60.11 to 192.168.60.254

4.3 Routing

Routing between terminals is handled by OSPF Area 0 over the 10.0.0.0/30 backbone. Each TIA router advertises its local VLANs and WAN link, keeps its own static default route, and shares that default into OSPF. This gives the airport a simple dual-ISP design without using banned features.

4.4 Security Architecture

Security starts with VLAN isolation and is strengthened by the guest ACLs INTL_GUEST_SECURE and DOM_GUEST_SECURE. These rules allow approved DNS traffic and web access to the public guest portal at 192.168.40.2 while

blocking private internal ranges. WPA2-PSK protects guest wireless access, and MGMT_ONLY protects router login access. Keeping the staff and guest portals on different server IPs makes the Packet Tracer policy cleaner and easier to verify.

4.5 Website Design

The CAAN website is kept simple so it works in Packet Tracer. The staff portal is the default index.html page on CAAN-Server and is reached through tia.caan.gov.np. The public guest portal is the default index.html page on DNS-Dom-Server and is reached through guest.tia.caan.gov.np. This split keeps the user experience simple and also lets the guest ACLs enforce access by destination IP.

5 Configurations

5.1 TIA-R1 Configuration (International Terminal Gateway)

TIA-R1 is the International terminal gateway. It provides Router-on-a-Stick gateways for VLANs 10, 20, and 30, relays DHCP requests to DHCP-Intl-Server, exchanges routes with TIA-R2, and applies the International guest ACL.

TIA-R1_Config.txt

```
Router> enable
Router# configure terminal
Router(config)# hostname TIA-R1

! Router-on-a-Stick base interface
TIA-R1(config)# interface GigabitEthernet0/0
TIA-R1(config-if)# no ip address
TIA-R1(config-if)# no shutdown
TIA-R1(config-if)# exit

! VLAN 10 - Immigration and Authority
TIA-R1(config)# interface GigabitEthernet0/0.10
TIA-R1(config-subif)# encapsulation dot1Q 10
TIA-R1(config-subif)# ip address 192.168.10.1 255.255.255.0
TIA-R1(config-subif)# ip helper-address 192.168.10.3
TIA-R1(config-subif)# exit

! VLAN 20 - International Airlines
TIA-R1(config)# interface GigabitEthernet0/0.20
TIA-R1(config-subif)# encapsulation dot1Q 20
TIA-R1(config-subif)# ip address 192.168.20.1 255.255.255.0
TIA-R1(config-subif)# ip helper-address 192.168.10.3
TIA-R1(config-subif)# exit

! VLAN 30 - International Guest Wi-Fi
TIA-R1(config)# interface GigabitEthernet0/0.30
TIA-R1(config-subif)# encapsulation dot1Q 30
TIA-R1(config-subif)# ip address 192.168.30.1 255.255.255.0
TIA-R1(config-subif)# ip helper-address 192.168.10.3
```

```
TIA-R1(config-subif)# exit

! Backbone link to TIA-R2
TIA-R1(config)# interface GigabitEthernet0/1
TIA-R1(config-if)# ip address 10.0.0.1 255.255.255.252
TIA-R1(config-if)# no shutdown
TIA-R1(config-if)# exit

! Nepal Telecom uplink
TIA-R1(config)# interface GigabitEthernet0/2
TIA-R1(config-if)# ip address 203.0.113.2 255.255.255.252
TIA-R1(config-if)# no shutdown
TIA-R1(config-if)# exit

! OSPF
TIA-R1(config)# router ospf 1
TIA-R1(config-router)# router-id 1.1.1.1
TIA-R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
TIA-R1(config-router)# network 192.168.20.0 0.0.0.255 area 0
TIA-R1(config-router)# network 192.168.30.0 0.0.0.255 area 0
TIA-R1(config-router)# network 10.0.0.0 0.0.0.3 area 0
TIA-R1(config-router)# network 203.0.113.0 0.0.0.3 area 0
TIA-R1(config-router)# default-information originate
TIA-R1(config-router)# exit

! Default route
TIA-R1(config)# ip route 0.0.0.0 0.0.0.0 203.0.113.1

! International guest security ACL
TIA-R1(config)# ip access-list extended INTL_GUEST_SECURE
TIA-R1(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 192.168.50.0
    0.0.0.255
TIA-R1(config-ext-nacl)# permit udp 192.168.30.0 0.0.0.255 host
    192.168.10.2 eq domain
TIA-R1(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 host
    192.168.40.2 eq www
TIA-R1(config-ext-nacl)# permit tcp 192.168.30.0 0.0.0.255 host
    192.168.40.2 eq 443
```

```
TIA-R1(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 192.168.0.0
    0.0.255.255
TIA-R1(config-ext-nacl)# permit ip any any
TIA-R1(config-ext-nacl)# exit

TIA-R1(config)# interface GigabitEthernet0/0.30
TIA-R1(config-subif)# ip access-group INTL_GUEST_SECURE in
TIA-R1(config-subif)# exit

! Router management protection
TIA-R1(config)# ip access-list standard MGMT_ONLY
TIA-R1(config-std-nacl)# permit 192.168.10.0 0.0.0.255
TIA-R1(config-std-nacl)# permit 192.168.40.0 0.0.0.255
TIA-R1(config-std-nacl)# exit

TIA-R1(config)# line vty 0 4
TIA-R1(config-line)# access-class MGMT_ONLY in
TIA-R1(config-line)# exit

TIA-R1(config)# end
TIA-R1# copy running-config startup-config
```

5.2 TIA-R2 Configuration (Domestic Terminal Gateway)

TIA-R2 is the Domestic terminal gateway. It provides gateways for VLANs 40, 50, and 60, relays DHCP to DHCP-Dom-Server, connects to Worldlink, and applies the Domestic guest ACL.

TIA-R2_Config.txt

```
Router> enable
Router# configure terminal
Router(config)# hostname TIA-R2

! Router-on-a-Stick base interface
TIA-R2(config)# interface GigabitEthernet0/0
TIA-R2(config-if)# no ip address
TIA-R2(config-if)# no shutdown
```

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.10.3
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.10.3
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
 ip helper-address 192.168.10.3
 ip access-group INTL_GUEST_SECURE in
!
interface GigabitEthernet0/1
 ip address 10.0.0.1 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 ip address 203.0.113.2 255.255.255.252
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.20.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 203.0.113.0 0.0.0.3 area 0
 default-information originate
!
router rip
!
ip classless
ip route 0.0.0.0 0.0.0.0 203.0.113.1
!
ip flow-export version 9
!
!
ip access-list standard MGMT_ONLY
 permit 192.168.10.0 0.0.0.255
 permit 192.168.40.0 0.0.0.255
ip access-list extended INTL_GUEST_SECURE
 deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
 permit udp 192.168.30.0 0.0.0.255 host 192.168.10.2 eq domain
 permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.2 eq www
 permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.2 eq 443
 deny ip 192.168.30.0 0.0.0.255 192.168.0.0 0.0.255.255
 permit ip any any
```

Figure 2: TIA-R1 configuration showing subinterfaces, DHCP relay, OSPF, and guest ACL settings.

```
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/1     10.0.0.1        YES manual up                    up
GigabitEthernet0/2     203.0.113.2     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
TIA-R1#
```

Figure 3: TIA-R1 interface summary after configuration.

```
TIA-R2(config-if)# exit

! VLAN 40 - Domestic Airlines
TIA-R2(config)# interface GigabitEthernet0/0.40
TIA-R2(config-subif)# encapsulation dot1Q 40
TIA-R2(config-subif)# ip address 192.168.40.1 255.255.255.0
TIA-R2(config-subif)# ip helper-address 192.168.40.3
TIA-R2(config-subif)# exit

! VLAN 50 - Domestic Guest Wi-Fi
TIA-R2(config)# interface GigabitEthernet0/0.50
TIA-R2(config-subif)# encapsulation dot1Q 50
TIA-R2(config-subif)# ip address 192.168.50.1 255.255.255.0
TIA-R2(config-subif)# ip helper-address 192.168.40.3
TIA-R2(config-subif)# exit

! VLAN 60 - ATC and TETRA CNS
TIA-R2(config)# interface GigabitEthernet0/0.60
TIA-R2(config-subif)# encapsulation dot1Q 60
TIA-R2(config-subif)# ip address 192.168.60.1 255.255.255.0
TIA-R2(config-subif)# ip helper-address 192.168.40.3
TIA-R2(config-subif)# exit

! Backbone link to TIA-R1
TIA-R2(config)# interface GigabitEthernet0/1
TIA-R2(config-if)# ip address 10.0.0.2 255.255.255.252
TIA-R2(config-if)# no shutdown
TIA-R2(config-if)# exit

! Worldlink uplink
```

```
TIA-R2(config)# interface GigabitEthernet0/2
TIA-R2(config-if)# ip address 198.51.100.2 255.255.255.252
TIA-R2(config-if)# no shutdown
TIA-R2(config-if)# exit

! OSPF
TIA-R2(config)# router ospf 1
TIA-R2(config-router)# router-id 2.2.2.2
TIA-R2(config-router)# network 192.168.40.0 0.0.0.255 area 0
TIA-R2(config-router)# network 192.168.50.0 0.0.0.255 area 0
TIA-R2(config-router)# network 192.168.60.0 0.0.0.255 area 0
TIA-R2(config-router)# network 10.0.0.0 0.0.0.3 area 0
TIA-R2(config-router)# network 198.51.100.0 0.0.0.3 area 0
TIA-R2(config-router)# default-information originate
TIA-R2(config-router)# exit

! Default route
TIA-R2(config)# ip route 0.0.0.0 0.0.0.0 198.51.100.1

! Domestic guest security ACL
TIA-R2(config)# ip access-list extended DOM_GUEST_SECURE
TIA-R2(config-ext-nacl)# deny ip 192.168.50.0 0.0.0.255 192.168.30.0
    0.0.0.255
TIA-R2(config-ext-nacl)# permit udp 192.168.50.0 0.0.0.255 host
    192.168.40.2 eq domain
TIA-R2(config-ext-nacl)# permit udp 192.168.50.0 0.0.0.255 host
    192.168.10.2 eq domain
TIA-R2(config-ext-nacl)# permit tcp 192.168.50.0 0.0.0.255 host
    192.168.40.2 eq www
TIA-R2(config-ext-nacl)# permit tcp 192.168.50.0 0.0.0.255 host
    192.168.40.2 eq 443
TIA-R2(config-ext-nacl)# deny ip 192.168.50.0 0.0.0.255 192.168.0.0
    0.0.255.255
TIA-R2(config-ext-nacl)# permit ip any any
TIA-R2(config-ext-nacl)# exit

TIA-R2(config)# interface GigabitEthernet0/0.50
TIA-R2(config-subif)# ip access-group DOM_GUEST_SECURE in
```

```
TIA-R2(config-subif)# exit

! Router management protection
TIA-R2(config)# ip access-list standard MGMT_ONLY
TIA-R2(config-std-nacl)# permit 192.168.10.0 0.0.0.255
TIA-R2(config-std-nacl)# permit 192.168.40.0 0.0.0.255
TIA-R2(config-std-nacl)# exit

TIA-R2(config)# line vty 0 4
TIA-R2(config-line)# access-class MGMT_ONLY in
TIA-R2(config-line)# exit

TIA-R2(config)# end
TIA-R2# copy running-config startup-config
```

5.3 How the Guest ACL Works

The guest ACLs follow a simple zero-trust logic:

1. Guests are allowed to use approved DNS servers so they can resolve the airport hostnames.
2. Guests are allowed to open only the public portal at guest.tia.caan.gov.np on 192.168.40.2.
3. Private airport address ranges are denied, so guests cannot reach staff, airline, or operational systems.
4. Remaining public traffic is still allowed, so guest users can reach non-private destinations.
5. Router login access is controlled separately by the MGMT_ONLY ACL on the VTY lines.

5.4 ISP Router Configuration

The two ISP routers stay outside the OSPF domain. Their job is to terminate the WAN links and provide static return routes back to all six airport VLANs.

```
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
 ip helper-address 192.168.40.3
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 192.168.50.1 255.255.255.0
 ip helper-address 192.168.40.3
 ip access-group DOM_GUEST_SECURE in
!
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 192.168.60.1 255.255.255.0
 ip helper-address 192.168.40.3
!
interface GigabitEthernet0/1
 ip address 10.0.0.2 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 ip address 198.51.100.2 255.255.255.252
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 router-id 2.2.2.2
 log-adjacency-changes
 network 192.168.40.0 0.0.0.255 area 0
 network 192.168.50.0 0.0.0.255 area 0
 network 192.168.60.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 198.51.100.0 0.0.0.3 area 0
 default-information originate
!
router rip
!
ip classless
ip route 0.0.0.0 0.0.0.0 198.51.100.1
!
ip flow-export version 9
!
!
ip access-list standard MGMT_ONLY
 permit 192.168.10.0 0.0.0.255
 permit 192.168.40.0 0.0.0.255
ip access-list extended DOM_GUEST_SECURE
 deny ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255
 permit udp 192.168.50.0 0.0.0.255 host 192.168.40.2 eq domain
 permit udp 192.168.50.0 0.0.0.255 host 192.168.10.2 eq domain
 permit tcp 192.168.50.0 0.0.0.255 host 192.168.40.2 eq www
 permit tcp 192.168.50.0 0.0.0.255 host 192.168.40.2 eq 443
 deny ip 192.168.50.0 0.0.0.255 192.168.0.0 0.0.255.255
```

Figure 4: TIA-R2 configuration showing subinterfaces, DHCP relay, OSPF, and guest ACL settings.

```
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/0.50  192.168.50.1    YES manual up                    up
GigabitEthernet0/0.60  192.168.60.1    YES manual up                    up
GigabitEthernet0/1     10.0.0.2        YES manual up                    up
GigabitEthernet0/2     198.51.100.2    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
TIA-R2#
```

Figure 5: TIA-R2 interface summary after configuration.

```
TIA-R1#show ip access-lists INTL_GUEST_SECURE
Extended IP access list INTL_GUEST_SECURE
    deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255 (4 match(es))
    permit udp 192.168.30.0 0.0.0.255 host 192.168.10.2 eq domain (3 match(es))
    permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.2 eq www (9 match(es))
    permit tcp 192.168.30.0 0.0.0.255 host 192.168.40.2 eq 443
    deny ip 192.168.30.0 0.0.0.255 192.168.0.0 0.0.255.255 (8 match(es))
    permit ip any any (4 match(es))
```

Figure 6: International guest ACL on TIA-R1 showing the applied rules.

```
TIA-R2#show ip access-lists DOM_GUEST_SECURE
Extended IP access list DOM_GUEST_SECURE
    deny ip 192.168.50.0 0.0.0.255 192.168.30.0 0.0.0.255
    permit udp 192.168.50.0 0.0.0.255 host 192.168.40.2 eq domain
    permit udp 192.168.50.0 0.0.0.255 host 192.168.10.2 eq domain
    permit tcp 192.168.50.0 0.0.0.255 host 192.168.40.2 eq www
    permit tcp 192.168.50.0 0.0.0.255 host 192.168.40.2 eq 443
    deny ip 192.168.50.0 0.0.0.255 192.168.0.0 0.0.255.255
    permit ip any any (4 match(es))
```

Figure 7: Domestic guest ACL on TIA-R2 showing the applied rules.

NT-Router_Config.txt

```
Router> enable
Router# configure terminal
Router(config)# hostname NT-Router

NT-Router(config)# interface GigabitEthernet0/0
NT-Router(config-if)# ip address 203.0.113.1 255.255.255.252
NT-Router(config-if)# no shutdown
NT-Router(config-if)# exit

! Return routes to all airport networks
NT-Router(config)# ip route 192.168.10.0 255.255.255.0 203.0.113.2
NT-Router(config)# ip route 192.168.20.0 255.255.255.0 203.0.113.2
NT-Router(config)# ip route 192.168.30.0 255.255.255.0 203.0.113.2
NT-Router(config)# ip route 192.168.40.0 255.255.255.0 203.0.113.2
NT-Router(config)# ip route 192.168.50.0 255.255.255.0 203.0.113.2
NT-Router(config)# ip route 192.168.60.0 255.255.255.0 203.0.113.2

NT-Router(config)# end
NT-Router# copy running-config startup-config
```

WL-Router_Config.txt

```
Router> enable
Router# configure terminal
Router(config)# hostname WL-Router

WL-Router(config)# interface GigabitEthernet0/0
WL-Router(config-if)# ip address 198.51.100.1 255.255.255.252
WL-Router(config-if)# no shutdown
WL-Router(config-if)# exit

! Return routes to all airport networks
WL-Router(config)# ip route 192.168.10.0 255.255.255.0 198.51.100.2
WL-Router(config)# ip route 192.168.20.0 255.255.255.0 198.51.100.2
WL-Router(config)# ip route 192.168.30.0 255.255.255.0 198.51.100.2
WL-Router(config)# ip route 192.168.40.0 255.255.255.0 198.51.100.2
WL-Router(config)# ip route 192.168.50.0 255.255.255.0 198.51.100.2
WL-Router(config)# ip route 192.168.60.0 255.255.255.0 198.51.100.2
```

```
17
18 WL-Router(config)# end
19 WL-Router# copy running-config startup-config
```

```
NT-Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.10.0/24 [1/0] via 203.0.113.2
S    192.168.20.0/24 [1/0] via 203.0.113.2
S    192.168.30.0/24 [1/0] via 203.0.113.2
S    192.168.40.0/24 [1/0] via 203.0.113.2
S    192.168.50.0/24 [1/0] via 203.0.113.2
S    192.168.60.0/24 [1/0] via 203.0.113.2
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/0
L       203.0.113.1/32 is directly connected, GigabitEthernet0/0
```

Figure 8: NT-Router routing table showing return routes to the airport networks.

```
WL-Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.10.0/24 [1/0] via 198.51.100.2
S    192.168.20.0/24 [1/0] via 198.51.100.2
S    192.168.30.0/24 [1/0] via 198.51.100.2
S    192.168.40.0/24 [1/0] via 198.51.100.2
S    192.168.50.0/24 [1/0] via 198.51.100.2
S    192.168.60.0/24 [1/0] via 198.51.100.2
     198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       198.51.100.0/30 is directly connected, GigabitEthernet0/0
L       198.51.100.1/32 is directly connected, GigabitEthernet0/0
```

Figure 9: WL-Router routing table showing return routes to the airport networks.

5.5 Server Configuration Summary

Unlike the older sample report, this design uses dedicated Server-PT devices for DHCP, DNS, and HTTP. The exact GUI values are summarized below so the report stays concise but still keeps the real configuration details.

Table 6: Static server settings

Server	IP address	Gateway	Main role
CAAN-Server	192.168.10.100	192.168.10.1	Staff HTTP portal
DNS-Server	192.168.10.2	192.168.10.1	Primary DNS server
DNS-Dom-Server	192.168.40.2	192.168.40.1	Domestic DNS server and guest HTTP portal
DHCP-Intl-Server	192.168.10.3	192.168.10.1	DHCP for VLANs 10, 20, and 30
DHCP-Dom-Server	192.168.40.3	192.168.40.1	DHCP for VLANs 40, 50, and 60

Table 7: Key DHCP, DNS, and HTTP values used on the servers

Device	Applied settings
DHCP-Intl-Server	Pools VLAN 10, VLAN 20, and VLAN 30; start IPs 192.168.10.11, 192.168.20.11, and 192.168.30.11; gateway of each VLAN set to .1; DNS set to 192.168.10.2.
DHCP-Dom-Server	Pools VLAN 40, VLAN 50, and VLAN 60; start IPs 192.168.40.11, 192.168.50.11, and 192.168.60.11; gateway of each VLAN set to .1; primary DNS 192.168.40.2 and secondary DNS 192.168.10.2.
DNS-Server	DNS service on; A records tia.caan.gov.np → 192.168.10.100 and guest.tia.caan.gov.np → 192.168.40.2.
DNS-Dom-Server	DNS service on with the same two A records; HTTP service also on to host the public guest portal.
CAAN-Server	HTTP service on with the staff portal as index.html; guests are not allowed to reach this page through the guest ACL.

5.6 Switch VLAN and Trunk Configuration

All eight switches were configured with VLAN creation, access-port assignment, and trunking. To keep the report close to the older sample style, the main switch commands are grouped below in two compact code windows.

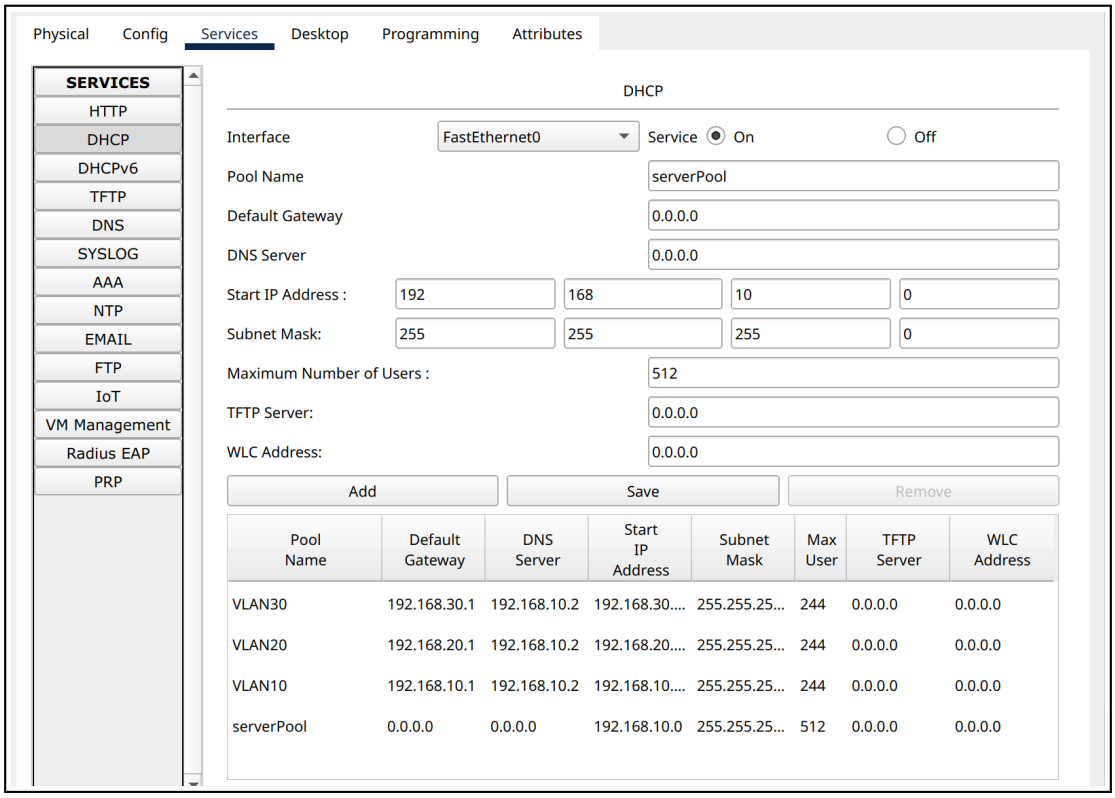


Figure 10: DHCP-Intl-Server pool configuration.

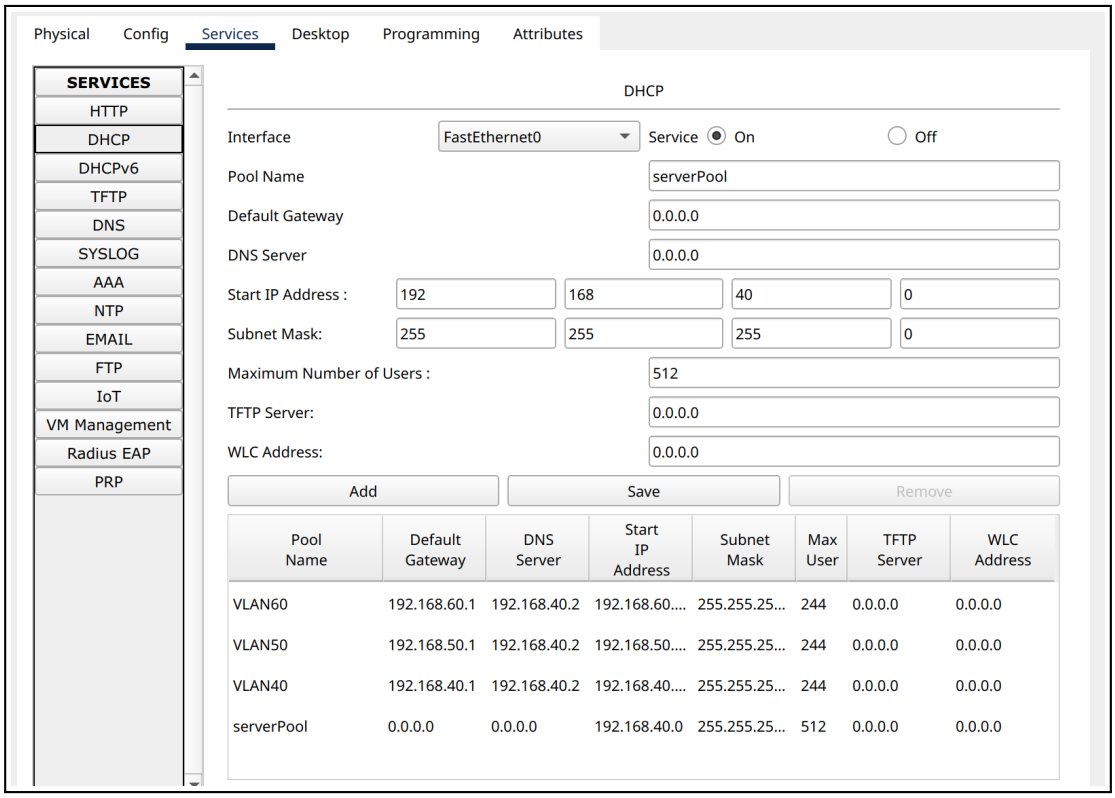


Figure 11: DHCP-Dom-Server pool configuration.

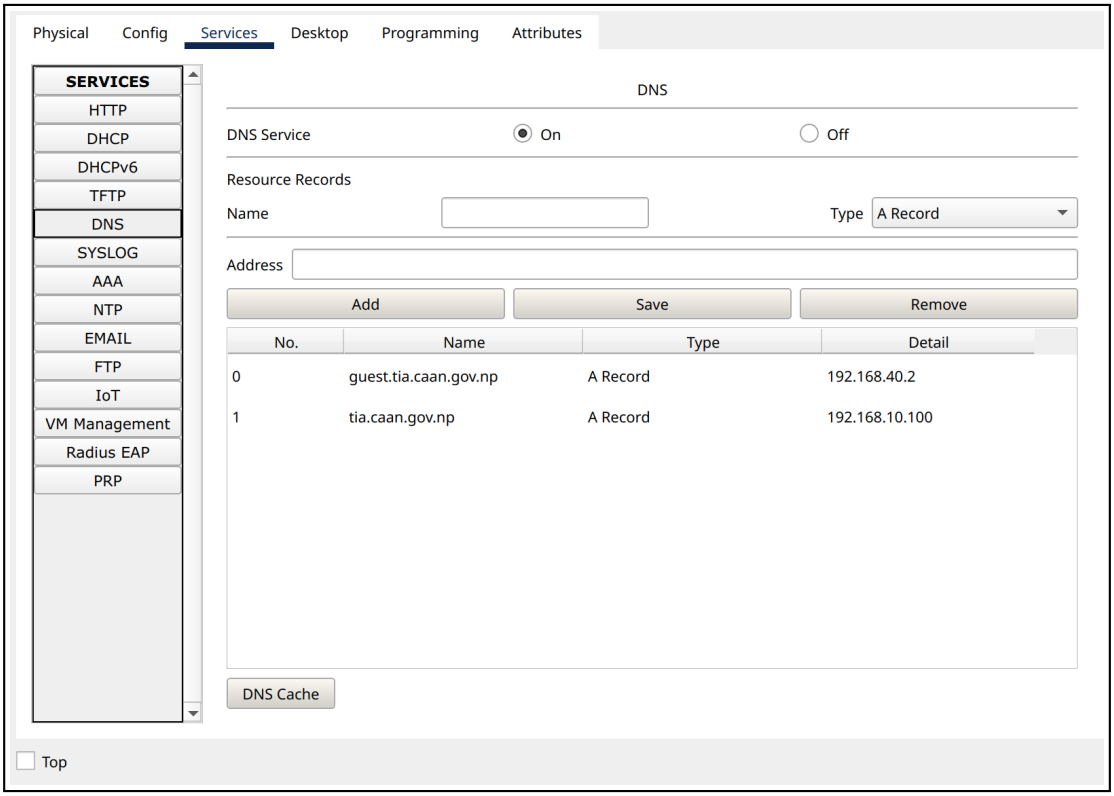


Figure 12: DNS-Server records for the staff and guest portals.

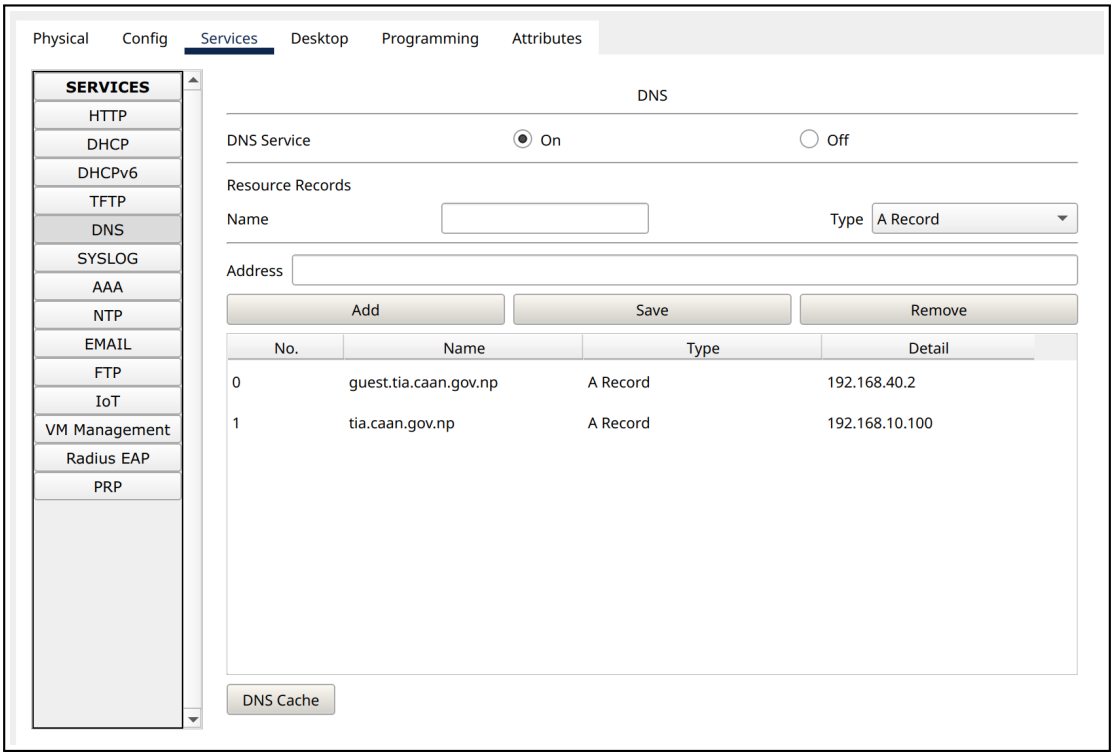


Figure 13: DNS-Dom-Server records matching the primary DNS service.

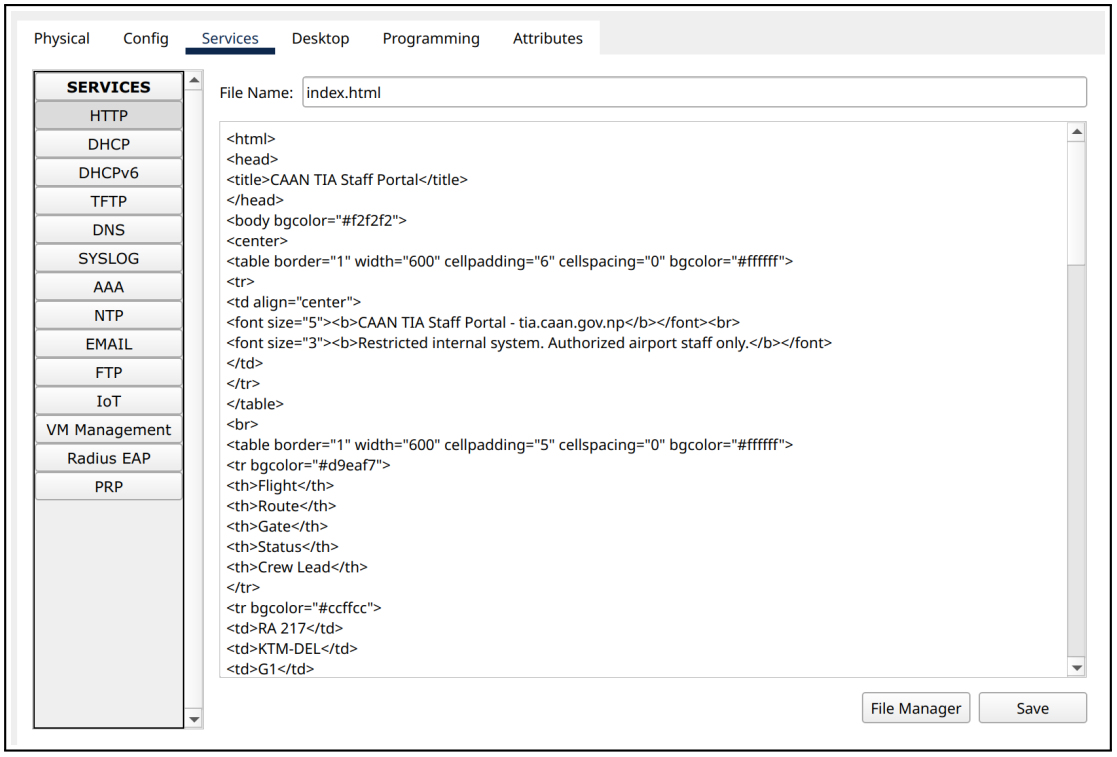


Figure 14: CAAN-Server HTTP page for trusted staff users.

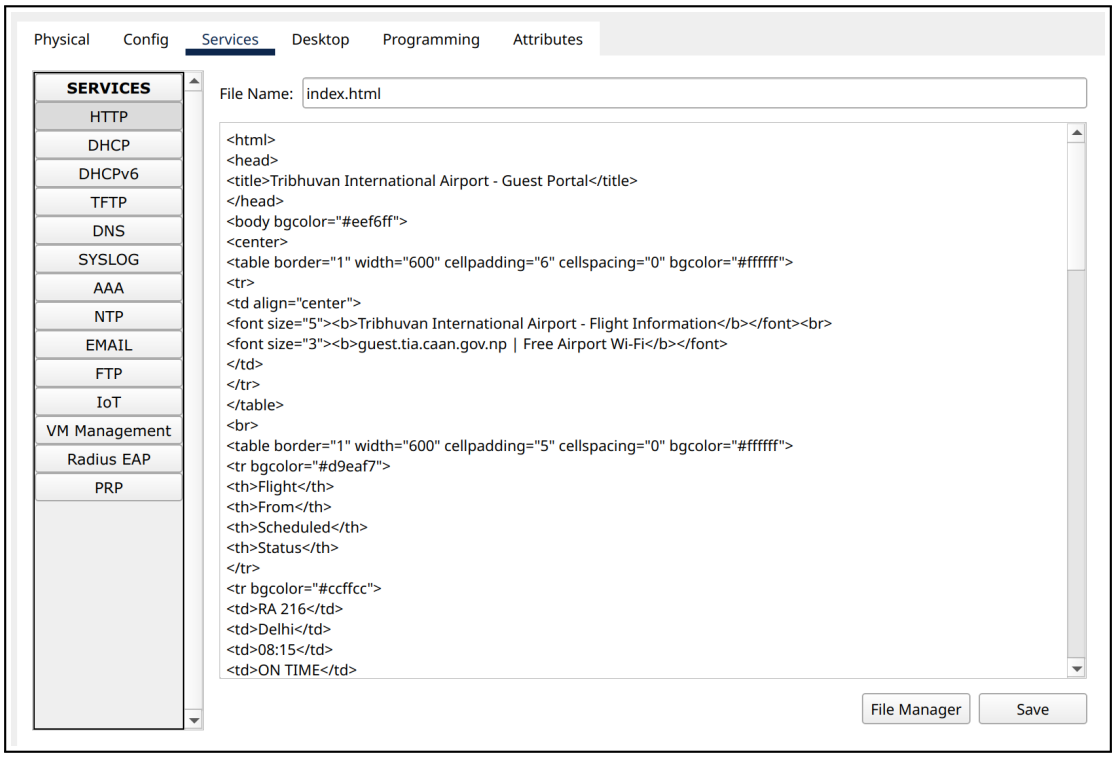


Figure 15: Public guest portal page hosted on DNS-Dom-Server.

International_Switches.txt

```
! Sw-Intl-Core
enable
configure terminal
hostname Sw-Intl-Core
vlan 10
 name Immigration_Authority
exit
vlan 20
 name International_Airlines
exit
vlan 30
 name Intl_Guest_WiFi
exit
interface gigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
exit
interface range fastEthernet0/22-24
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
exit
end
write memory

! Sw-Intl-Immig
enable
configure terminal
hostname Sw-Intl-Immig
vlan 10
 name Immigration_Authority
exit
vlan 20
 name International_Airlines
exit
vlan 30
 name Intl_Guest_WiFi
exit
```

```
interface range fastEthernet0/1-5
 switchport mode access
 switchport access vlan 10
exit
interface fastEthernet0/24
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
exit
end
write memory

! Sw-Intl-Airlines
enable
configure terminal
hostname Sw-Intl-Airlines
vlan 10
 name Immigration_Authority
exit
vlan 20
 name International_Airlines
exit
vlan 30
 name Intl_Guest_WiFi
exit
interface range fastEthernet0/1-3
 switchport mode access
 switchport access vlan 20
exit
interface fastEthernet0/24
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
exit
end
write memory

! Sw-Intl-Guest
enable
configure terminal
```

```
hostname Sw-Intl-Guest
vlan 10
 name Immigration_Authority
exit
vlan 20
 name International_Airlines
exit
vlan 30
 name Intl_Guest_WiFi
exit
interface range fastEthernet0/1-5
 switchport mode access
 switchport access vlan 30
exit
interface fastEthernet0/24
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
exit
end
write memory
```

Domestic_Switches.txt

```
! Sw-Dom-Core
enable
configure terminal
hostname Sw-Dom-Core
vlan 40
 name Domestic_Airlines
exit
vlan 50
 name Domestic_Guest_WiFi
exit
vlan 60
 name ATC_TETRA_CNS
exit
interface gigabitEthernet0/1
 switchport mode trunk
```

```
 switchport trunk allowed vlan 40,50,60
exit
interface range fastEthernet0/22-24
 switchport mode trunk
 switchport trunk allowed vlan 40,50,60
exit
end
write memory

! Sw-Dom-Airlines
enable
configure terminal
hostname Sw-Dom-Airlines
vlan 40
 name Domestic_Airlines
exit
vlan 50
 name Domestic_Guest_WiFi
exit
vlan 60
 name ATC_TETRA_CNS
exit
interface range fastEthernet0/1-5
 switchport mode access
 switchport access vlan 40
exit
interface fastEthernet0/24
 switchport mode trunk
 switchport trunk allowed vlan 40,50,60
exit
end
write memory

! Sw-Dom-Guest
enable
configure terminal
hostname Sw-Dom-Guest
vlan 40
```

```
 name Domestic_Airlines
exit
vlan 50
 name Domestic_Guest_WiFi
exit
vlan 60
 name ATC_TETRA_CNS
exit
interface range fastEthernet0/1-5
 switchport mode access
 switchport access vlan 50
exit
interface fastEthernet0/24
 switchport mode trunk
 switchport trunk allowed vlan 40,50,60
exit
end
write memory

! Sw-Dom-ATC
enable
configure terminal
hostname Sw-Dom-ATC
vlan 40
 name Domestic_Airlines
exit
vlan 50
 name Domestic_Guest_WiFi
exit
vlan 60
 name ATC_TETRA_CNS
exit
interface range fastEthernet0/1-3
 switchport mode access
 switchport access vlan 60
exit
interface fastEthernet0/24
 switchport mode trunk
```

```
 switchport trunk allowed vlan 40,50,60
exit
end
write memory
```

```
Sw-Intl-Core#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/22      on           802.1q         trunking      1
Fa0/23      on           802.1q         trunking      1
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/22      10,20,30
Fa0/23      10,20,30
Fa0/24      10,20,30

Port        Vlans allowed and active in management domain
Fa0/22      10,20,30
Fa0/23      10,20,30
Fa0/24      10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/22      10,20,30
Fa0/23      10,20,30
Fa0/24      10,20,30
```

Figure 16: International switch trunk verification.

5.7 End Device Verification

The final checks confirm that addressing, web access, and guest isolation work from the user side. Staff devices should receive correct leases and open the staff portal, while guest devices should receive guest addresses, fail to reach private systems, and open only the public guest page.

```
Sw-Dom-Core#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/22      on           802.1q         trunking      1
Fa0/23      on           802.1q         trunking      1
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/22      40,50,60
Fa0/23      40,50,60
Fa0/24      40,50,60

Port        Vlans allowed and active in management domain
Fa0/22      40,50,60
Fa0/23      40,50,60
Fa0/24      40,50,60

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/22      40,50,60
Fa0/23      40,50,60
Fa0/24      40,50,60
```

Figure 17: Domestic switch trunk verification.

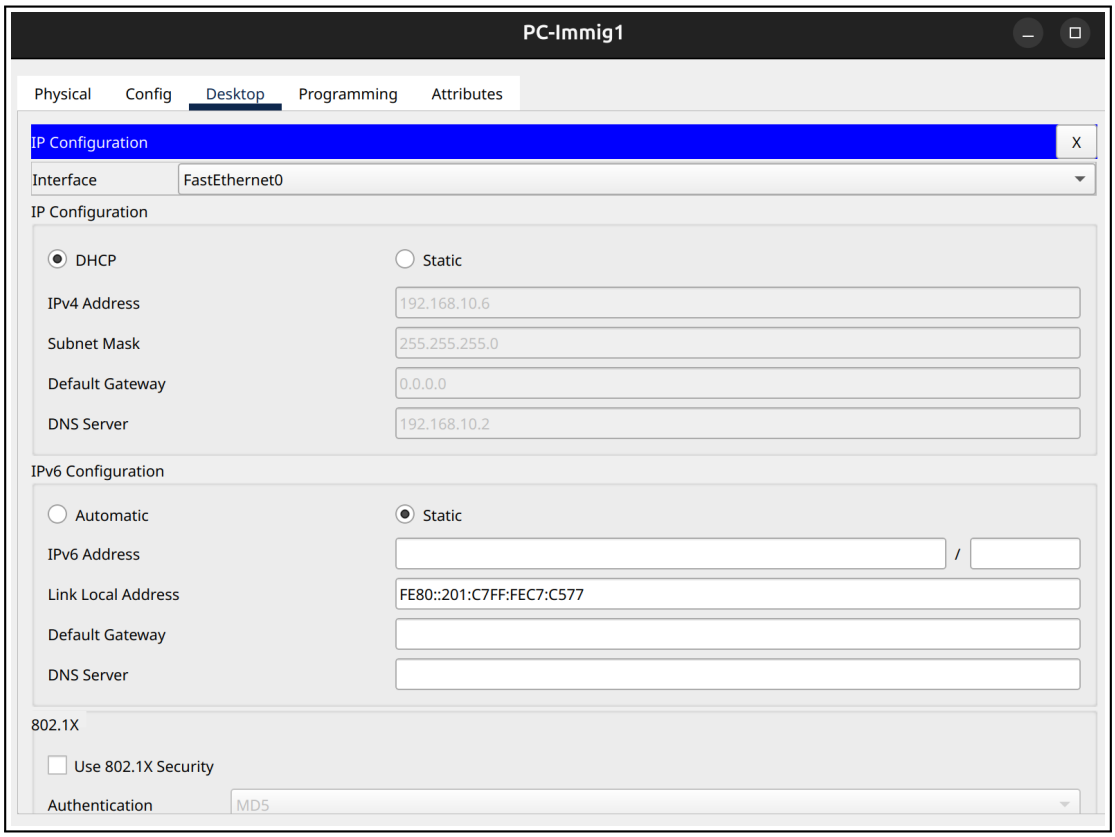


Figure 18: VLAN 10 PC receiving IP settings through DHCP.

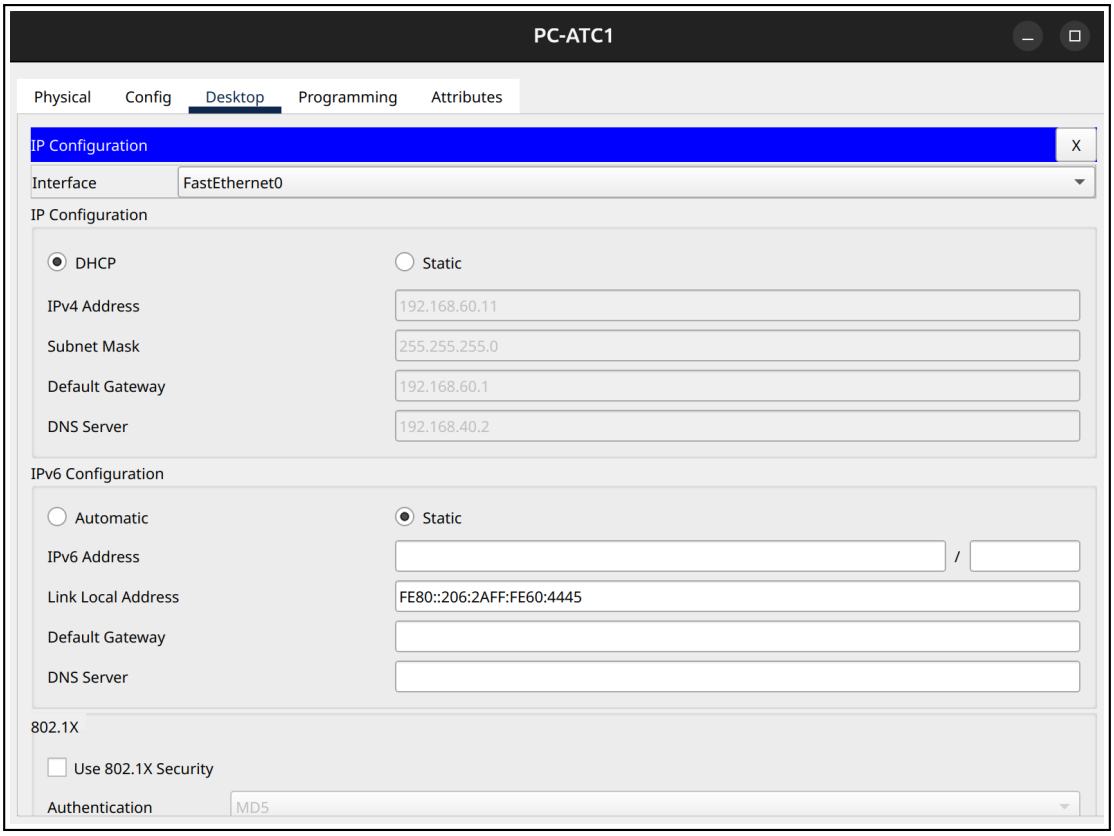


Figure 19: VLAN 60 PC receiving IP settings through DHCP.

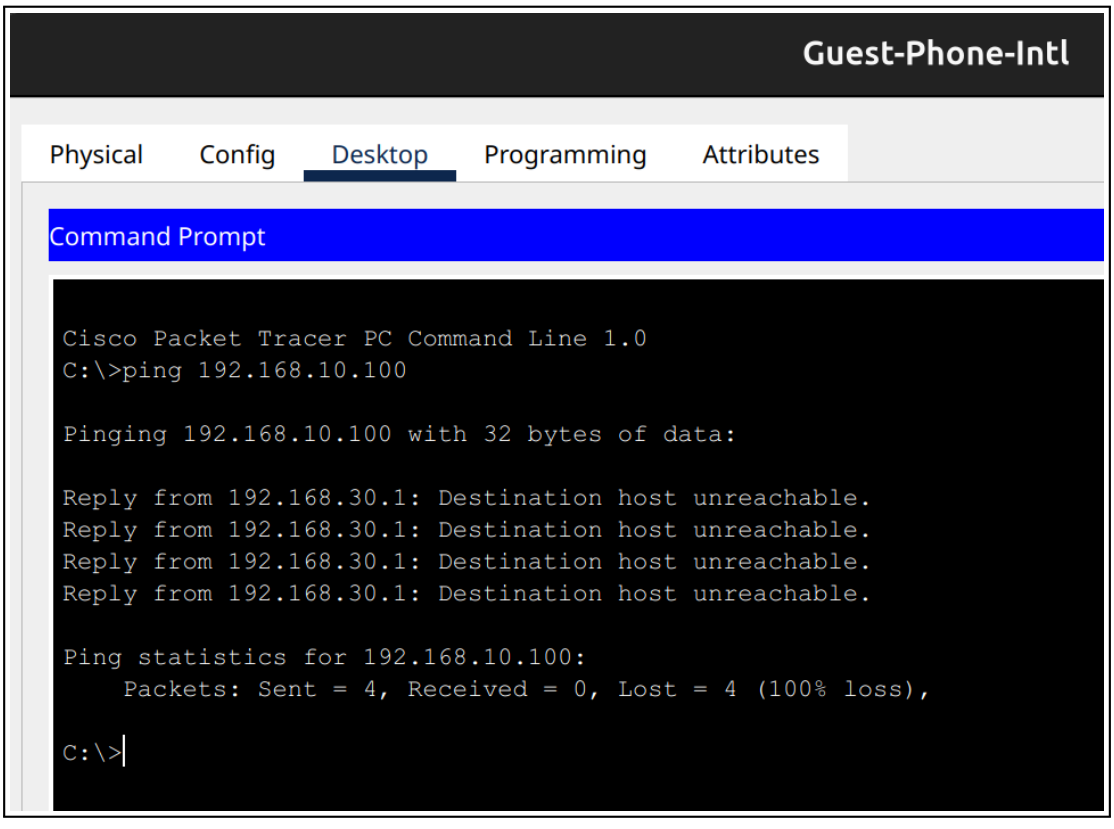


Figure 20: Guest device unable to reach an internal private host.

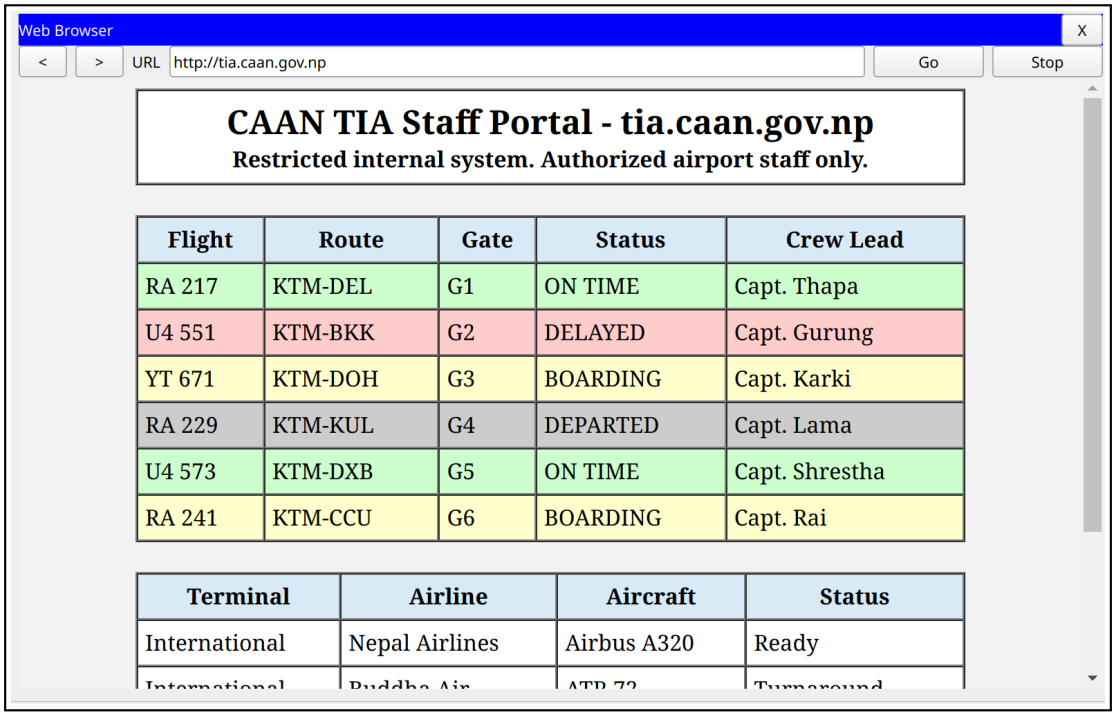


Figure 21: Staff user opening the internal portal through tia.caan.gov.np.

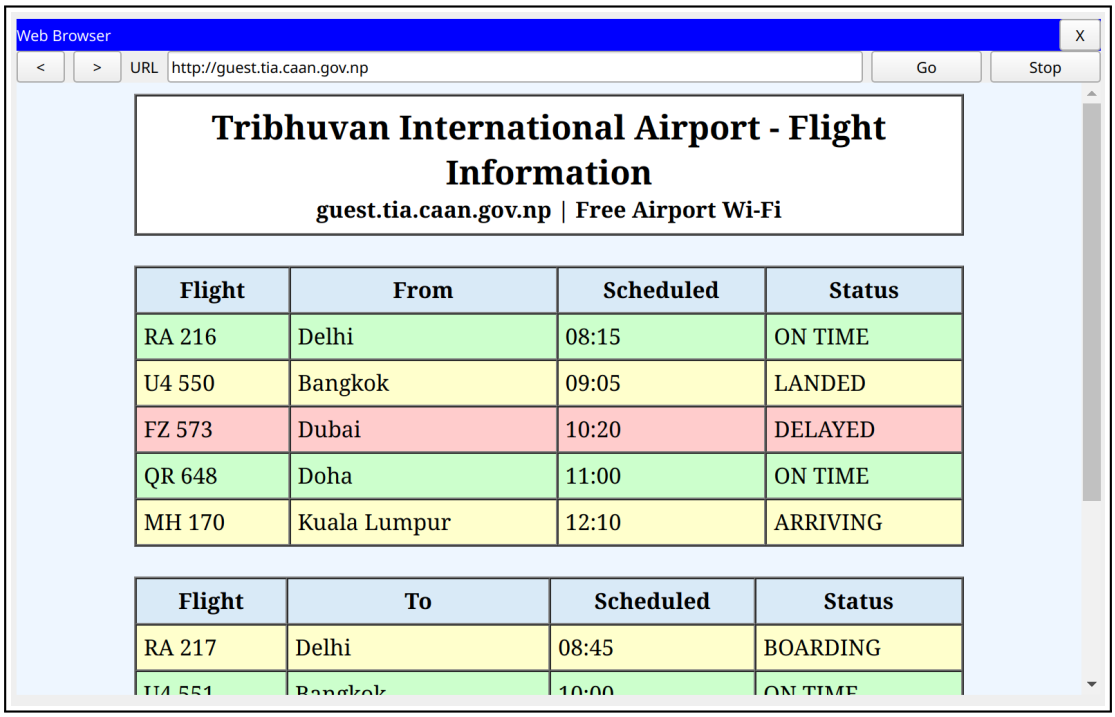


Figure 22: Guest user opening the public flight page through guest.tia.caan.gov.np.

6 Conclusion

This project built a dual-terminal airport network in Cisco Packet Tracer using only the technologies allowed by the course. The final design separates users into six VLANs, routes traffic with Router-on-a-Stick, and connects the two terminals through OSPF. DHCP, DNS, and HTTP are provided by Server-PT devices, and each terminal has its own ISP-facing default route. Guest users are controlled with extended ACLs, while router management is limited to trusted subnets through the MGMT_ONLY VTY rule. The design also improves resilience by keeping a Domestic-side DNS server and a separate public guest web host.

The completed topology shows that segmentation, routing, naming, web access, and policy control can work together in one coherent airport case study. Staff users can reach the internal portal, guest users can reach the public flight page through a separate hostname, and both sides can resolve the needed DNS records through the published design. Packet Tracer still has limits, because the web pages are basic and the security model is simpler than a real airport network. In a production environment, the design would need stronger security and resilience, such as redundant links, firewall zones, a DMZ for public services, RADIUS-based wireless authentication, centralized monitoring, backup services, and stronger failover design. Even with those limits, the case study meets its goal: it presents a clear, testable airport network that combines segmentation, routing, core services, and guest security in one practical design.